

Product Carbon Footprint Study Report

Product Name: Organic cotton hooded sweatshirt

Producer Name: ■■■

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: ■■■■

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: Organic cotton hooded sweatshirt

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

This study has two intended goals. The first one is to identify the stages of the life cycle of the hooded sweater that contribute the most to its environmental impacts. The second one is to assess the variability of those impacts depending on the production site or the behavior of the user. This LCA is carried as an internal request from GreenDelta, to provide openLCA users with a guided case LCA study.

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 organic cotton 2XL hooded sweater, with a weight of 750g, used for 1 year..

2. System Boundary

The product is assessed in a cradle-to-grave system, from the extraction of raw materials to the disposal of the used product in 2025. The extraction of raw materials and pre-processing consist of the production of the cotton fabric, the zipper, and polyester resin. This life cycle stage, as well as the manufacturing stage, take place in India. The product is then exported to Berlin where it is used and disposed.

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: ■■■■

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of Organic cotton hooded sweatshirt produced by ■■■■, per functional unit of product, from raw material acquisition to end of life is **20.800 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

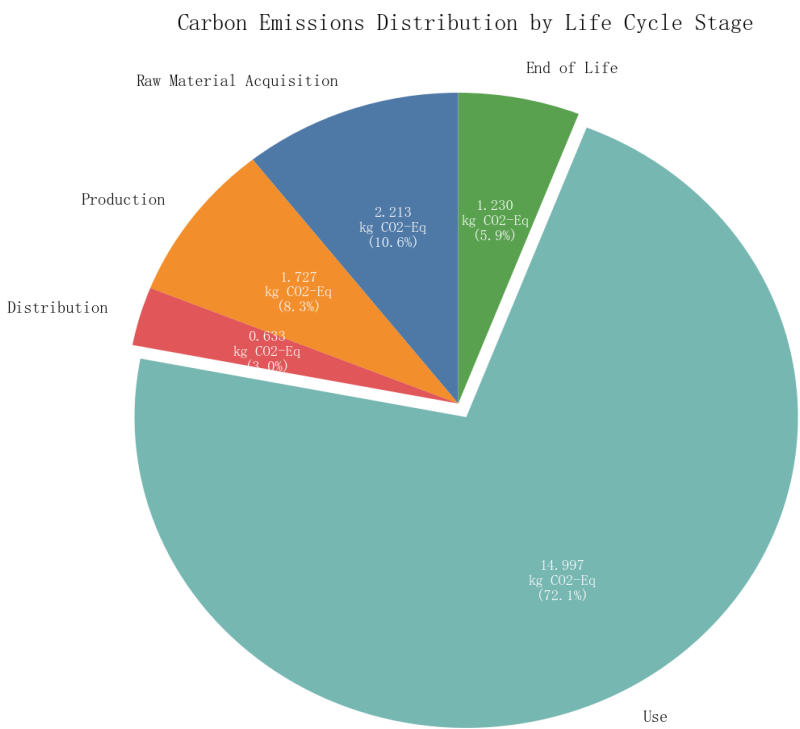


Table 1: Carbon Emissions by Life Cycle Stage

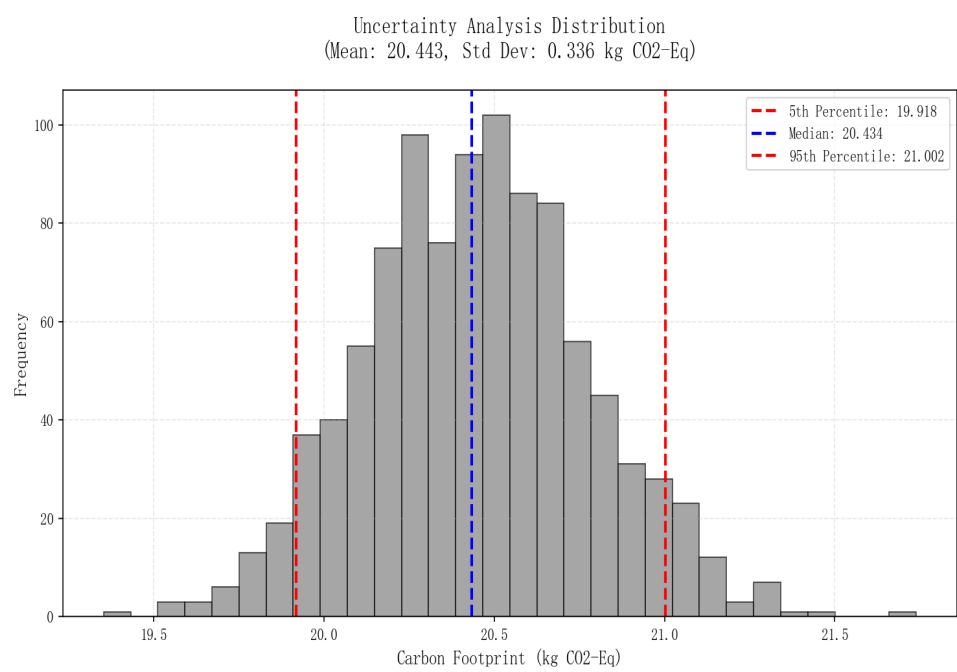
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	2.213	10.64
Production	1.727	8.30
Distribution	0.633	3.04

Use	14.997	72.10
End of Life	1.230	5.91
Total	20.800	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	20.443
Standard Deviation	0.336
Coefficient of Variation (CV)	1.65%
Median	20.434
5th Percentile	19.918
95th Percentile	21.002

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the organic cotton hooded sweatshirt is 20.800 kg CO₂-Eq, a value that is moderate for a garment of this type, reflecting the influence of both agricultural and consumer phases. The lifecycle stage analysis reveals a highly uneven contribution, with the use phase dominating at 72.1% (14.997 kg CO₂-Eq). This suggests that consumer behavior, particularly washing, drying, and ironing, drives the majority of emissions. The raw material stage accounts for 10.6% (2.213 kg CO₂-Eq), likely stemming from organic cotton cultivation and fiber processing, while production contributes 8.3% (1.727 kg CO₂-Eq). Distribution and end-of-life stages are minor, at 3.0% and 5.9% respectively. The major emission source is unequivocally the use phase, which overshadows all other stages. This indicates that efforts to reduce the product's carbon footprint must prioritize consumer usage patterns, as improvements in production or raw materials alone would have limited impact.

2. Statement of Assumptions and Limitations

Key assumptions in this study include that the organic cotton is grown without synthetic fertilizers and pesticides, which may lower raw material emissions but does not account for potential land-use change or irrigation energy. The use phase assumes a typical consumer washing and drying regimen (e.g., machine washing at 40°C and tumble drying), which may overestimate emissions for cold-wash or air-dry users. Data quality limitations include reliance on generic LCA databases for energy grids and textile processing, which may not reflect the specific producer's location or technology. The top process contributions are listed as unknown, indicating incomplete or missing data for disaggregated processes, reducing transparency. System boundary restrictions exclude capital goods (e.g., factory construction) and upstream packaging, which could add minor emissions. Additionally, the end-of-life stage assumes a specific disposal scenario (e.g., incineration or landfill), not accounting for recycling or reuse pathways that could alter results.

3. Suggestions for Improvement

- ****Target the use phase:**** Encourage consumers to wash in cold water (30°C or lower), air-dry instead of tumble-drying, and reduce washing frequency. Provide care labels with clear carbon-saving tips and partner with retailers to promote sustainable laundry habits.
- ****Improve raw material efficiency:**** Source organic cotton from farms using rain-fed irrigation and renewable energy for processing. Consider blending with recycled fibers to lower upstream emissions without compromising organic certification.
- ****Optimize production:**** Transition to renewable electricity in textile mills and garment factories. Implement lean manufacturing to reduce waste and energy use during cutting and sewing operations.
- ****Enhance data quality:**** Replace unknown process contributions with actual supplier data by conducting supplier-specific audits. Use primary data for energy and material inputs to improve accuracy and reduce uncertainty (current CV is low at 1.65%, but transparency is lacking).

- ****Extend product lifespan:**** Design for durability and repairability, and offer take-back programs for recycling or upcycling at end-of-life to reduce per-use carbon footprint over a longer lifespan.